Introduction

DS 5110: Big Data Systems

Summer 2026

Lecture 1a

Yue Cheng

UNIVERSITY of VIRGINIA

Introduction – Yue Cheng

- On the faculty of Data Science & Computer Science
  - Web: https://tddg.github.io
  - Email: mrz7dp@virginia.edu

- Current research: Designing **better AI systems**
  - Sustainable AI model storage
  - GPU-efficient Jupyter notebooks
  - Sustainable infra for generative content
  - Faster LLM fine-tuning / serving
  - …

Course staff and getting help

- Instructor: Yue Cheng
    - Office hours: Thursday 3pm – 4pm, Zoom

- GTAs:
    - Yuyang Cheng
    - Email: jrm9ga@virginia.edu
    - Office hours: Tuesday 2pm – 5pm, Zoom

    - Parvati Viswanathan
    - Email: bsg5sec@virginia.edu
    - Office hours: Wednesday 2pm – 5pm, Zoom

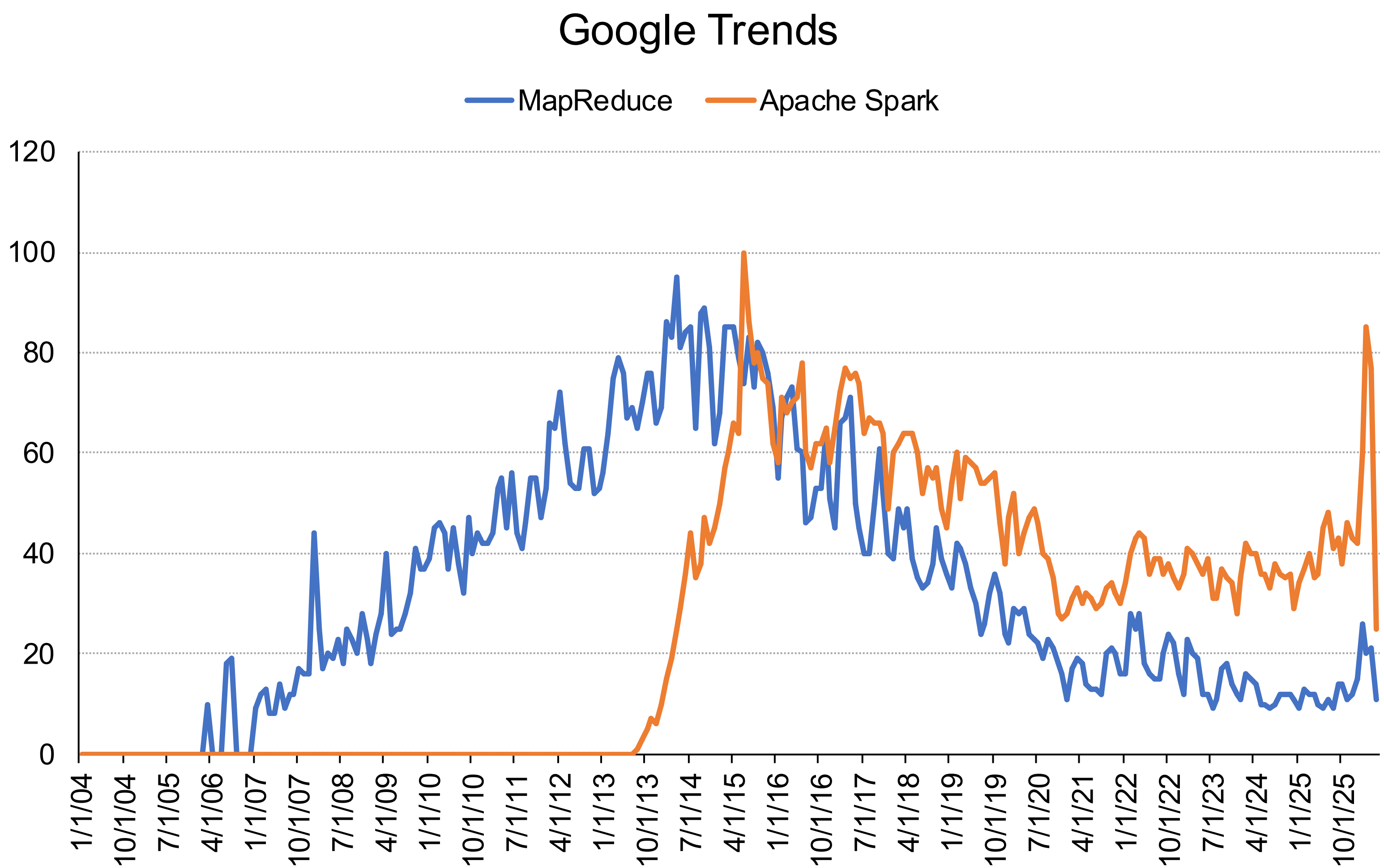
Course staff and getting help

- Offline discussion, questions: Ed
  - https://edstem.org/us/dashboard
  - Alternative place to ask questions about assignments, materials, and ideas
  - No anonymous posts or questions
  - Can use private posts to instructor/GTA
  - We are monitoring Ed several times a day
  - We will respond to questions in a batch manner

Today's agenda

- What is this course about?
- What will you do in this course?
- Shell, TUI, and AI coding CLI

A brief history about Big Data



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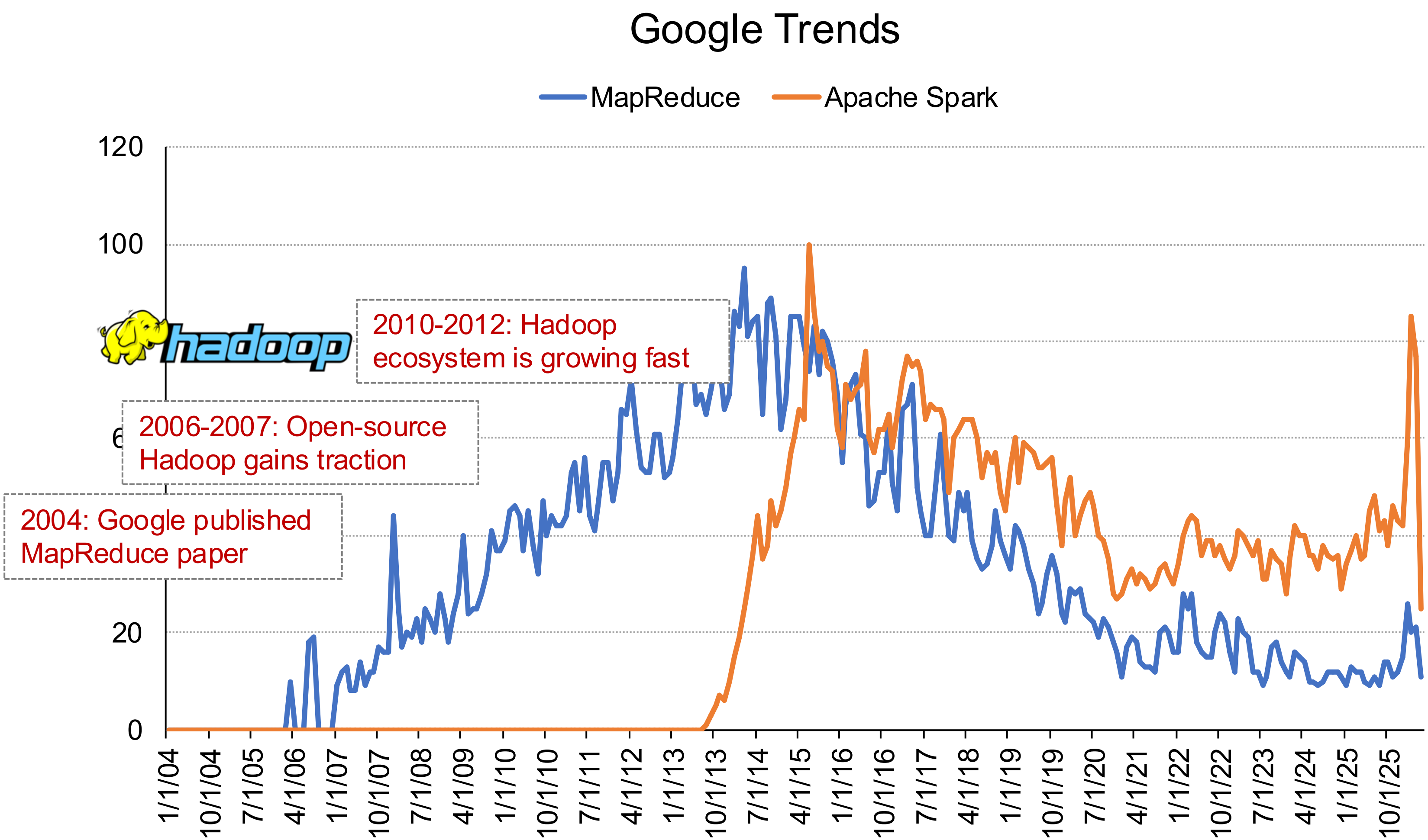

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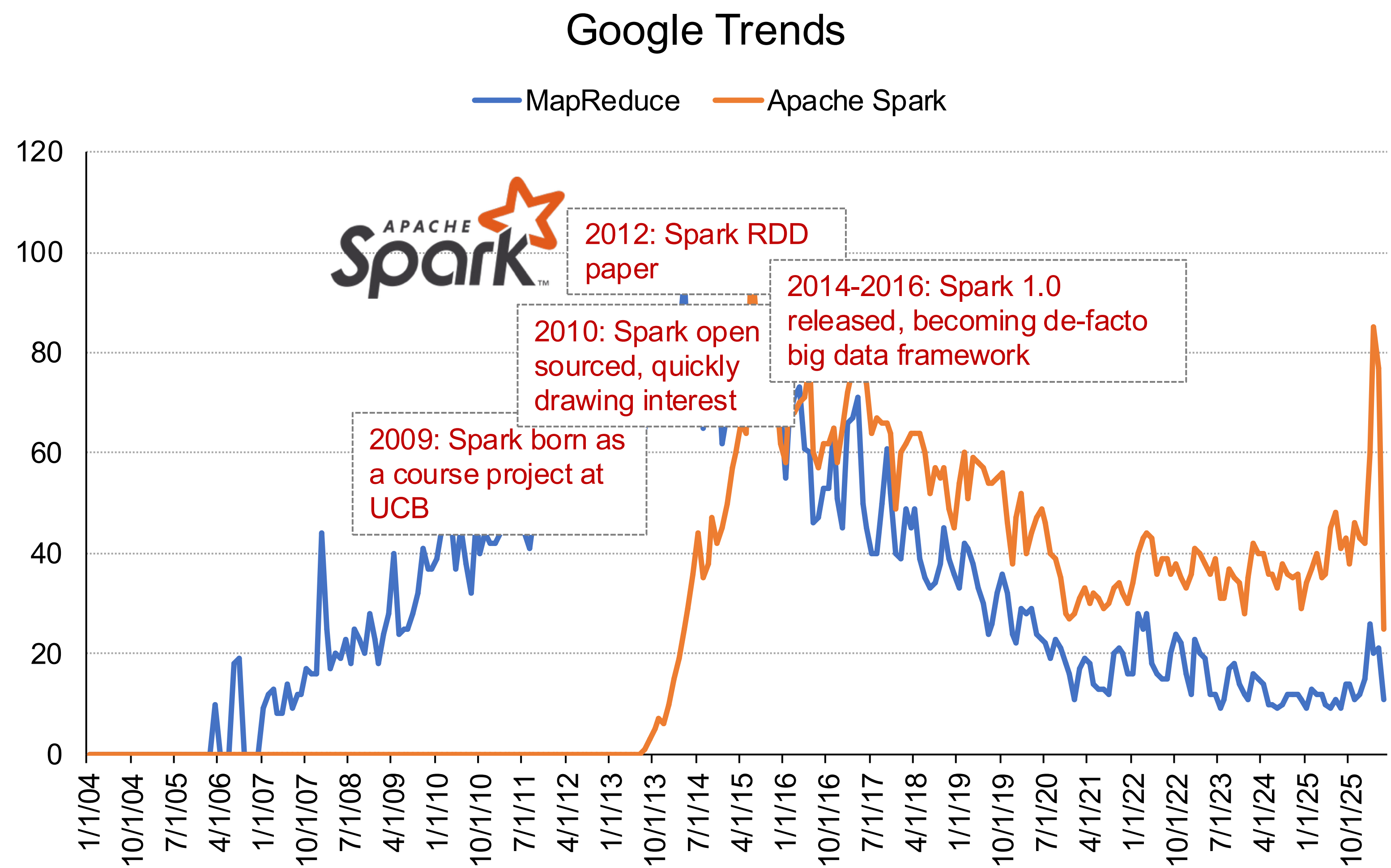

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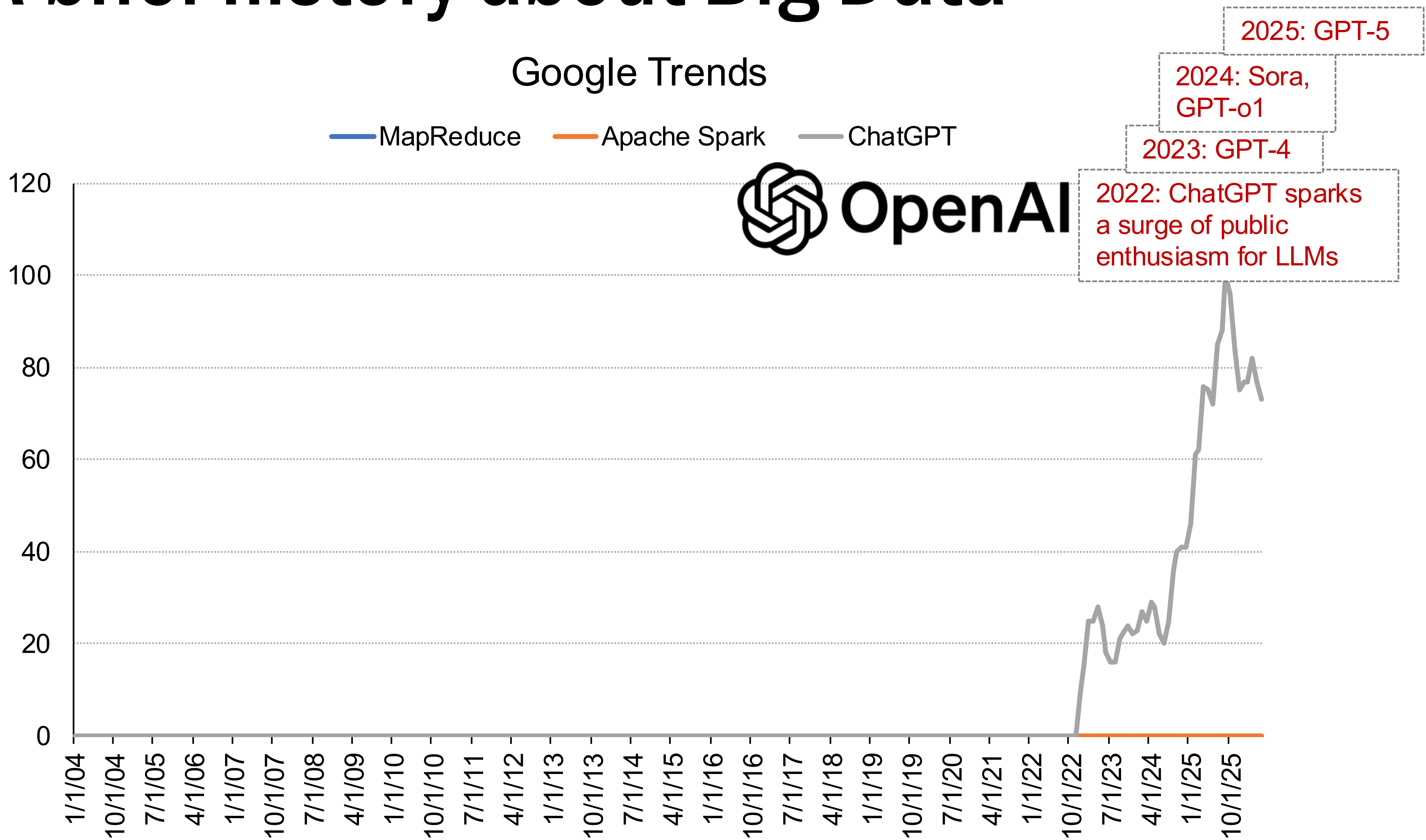

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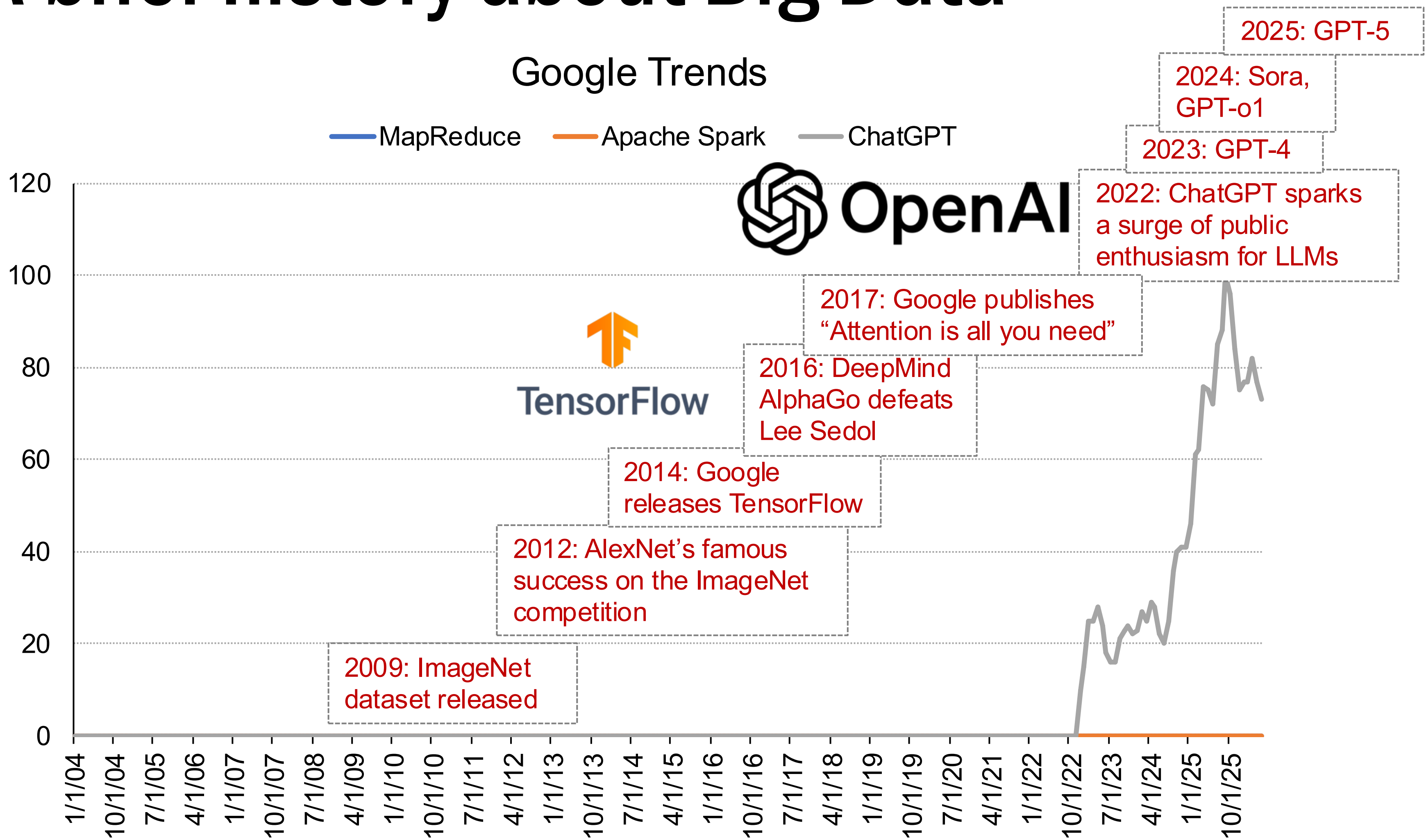
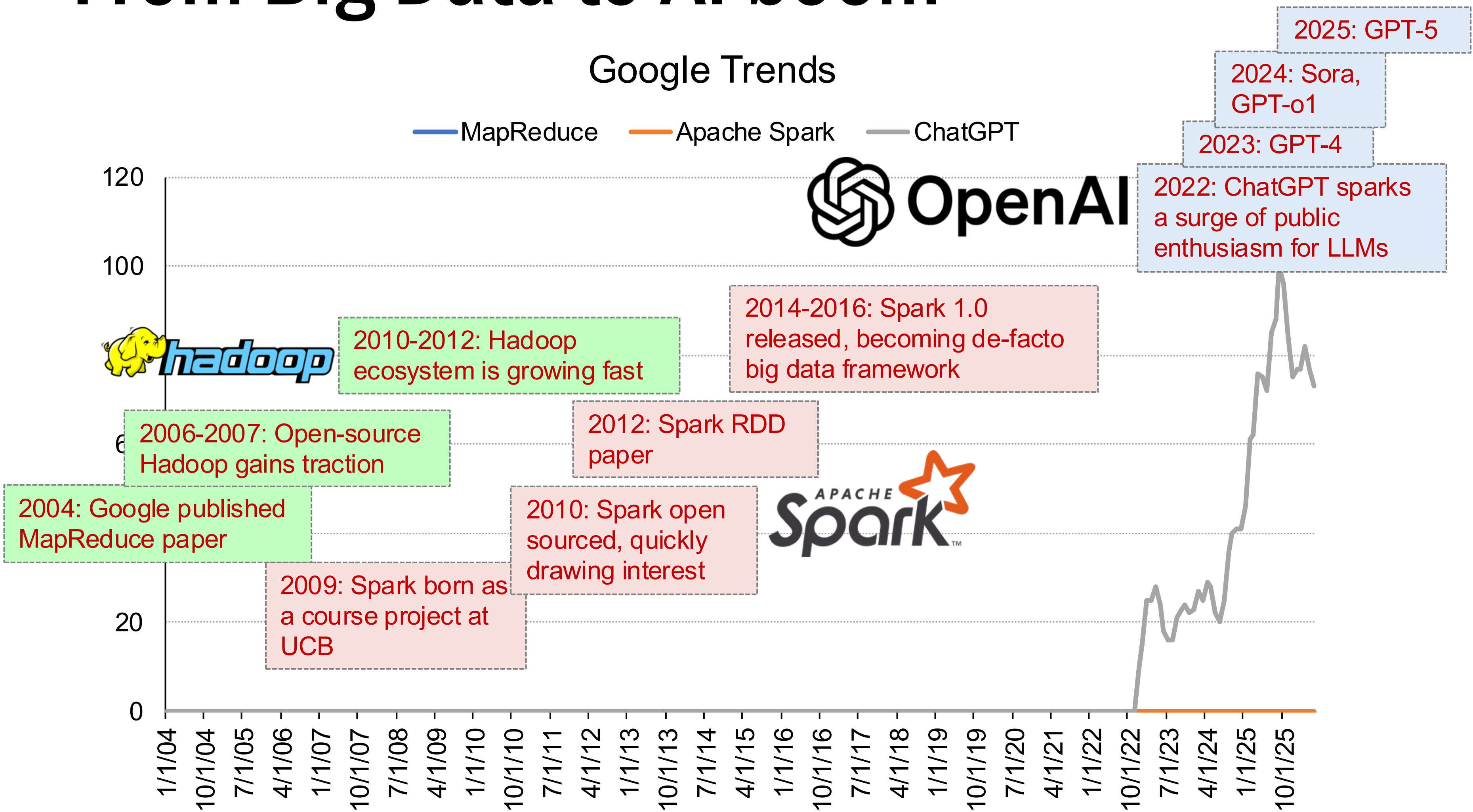


From Big Data to AI boom



Google circa 1997

Everything is about data

"… Storage space must be used efficiently to store indices and, optionally, the documents themselves. The indexing system must process hundreds of gigabytes of data efficiently…"

"The system… downloading the last 11 million pages in just 63 hours… The sorter can be run completely in parallel; using four machines, the whole process of sorting takes about 24 hours…"

The anatomy of a large-scale hypertextual Web search engine [1]

Sergey Brin [2], Lawrence Page [*,2]

Computer Science Department, Stanford University, Stanford, CA 94305, USA

Abstract

Google circa 2000

Commodity CPUs

Lots of disks

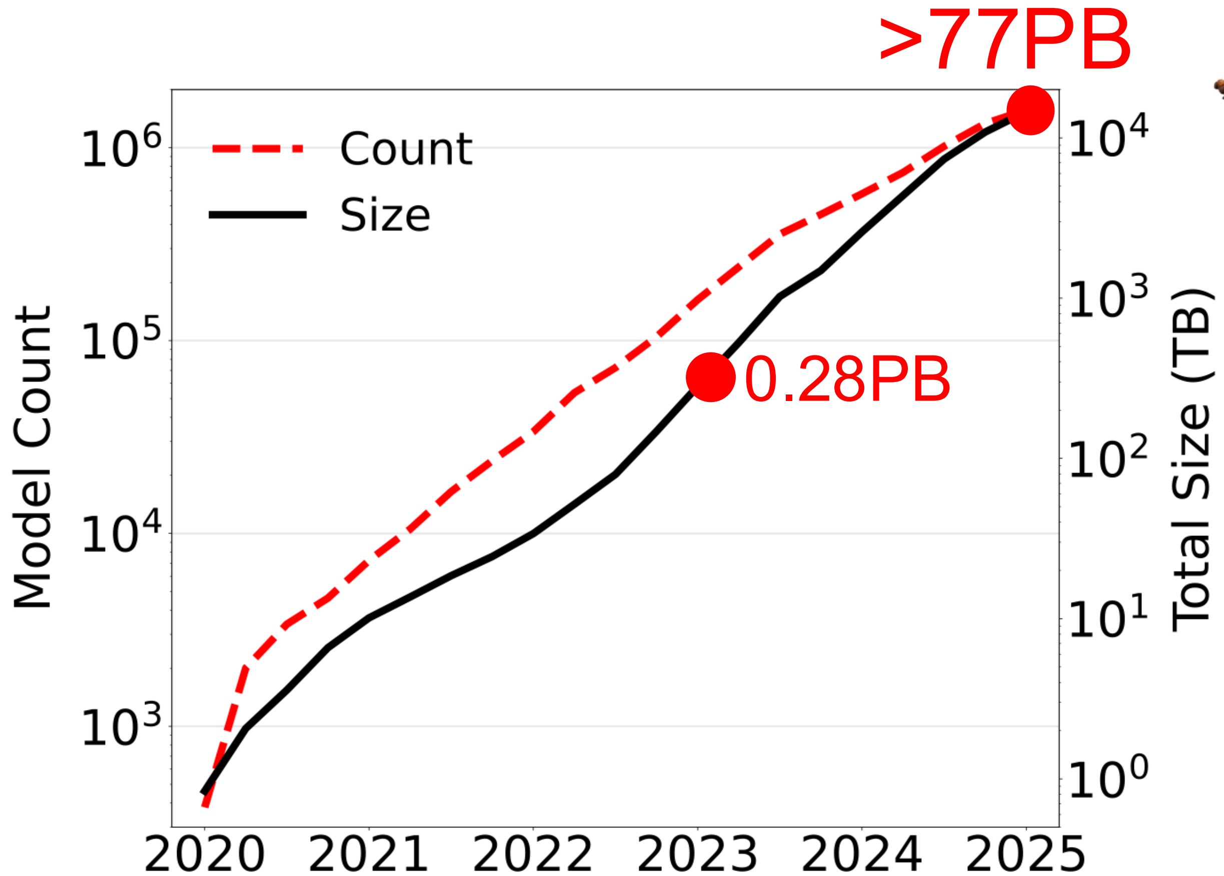
Low bandwidth network

Cheap!


Google

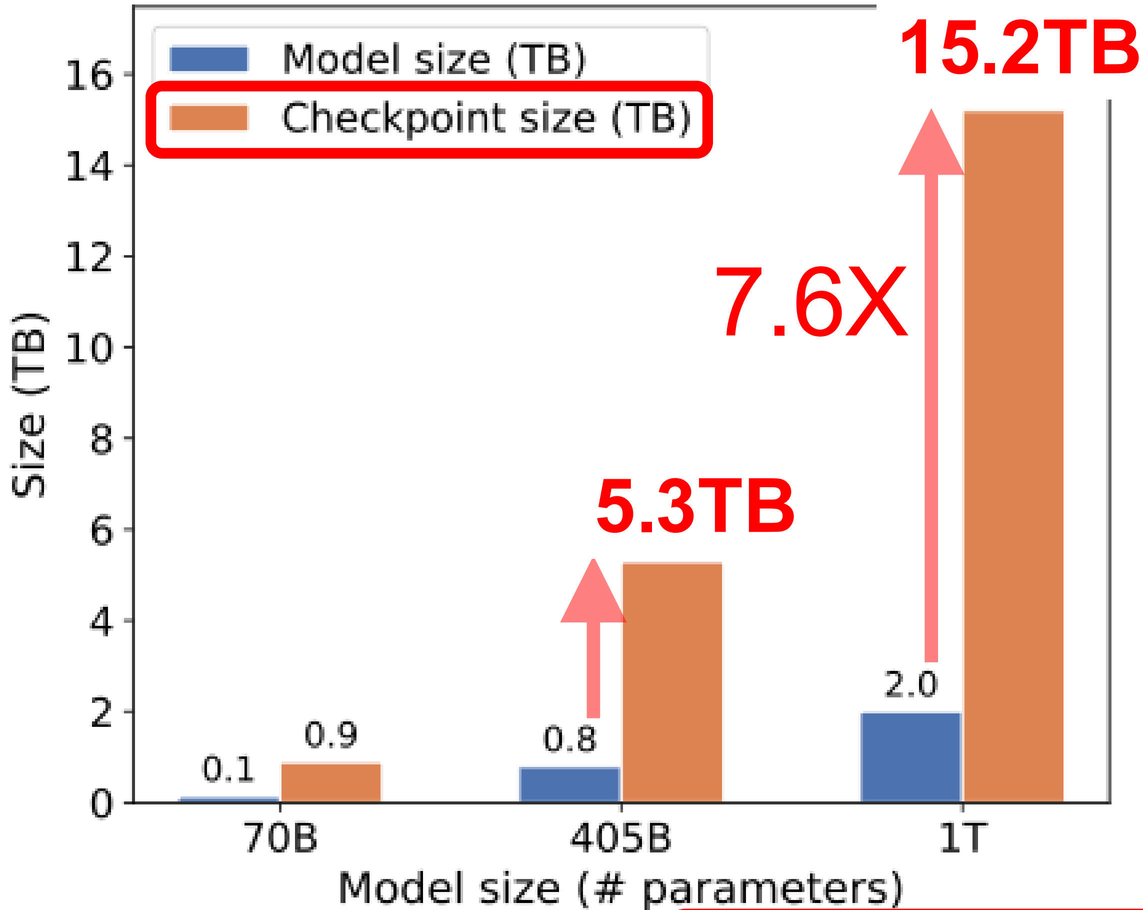
A Google Datacenter in Hamina, Finland

Data explosion in GenAI era



Hugging Face AI model hub

LLM training checkpoints are gigantic!

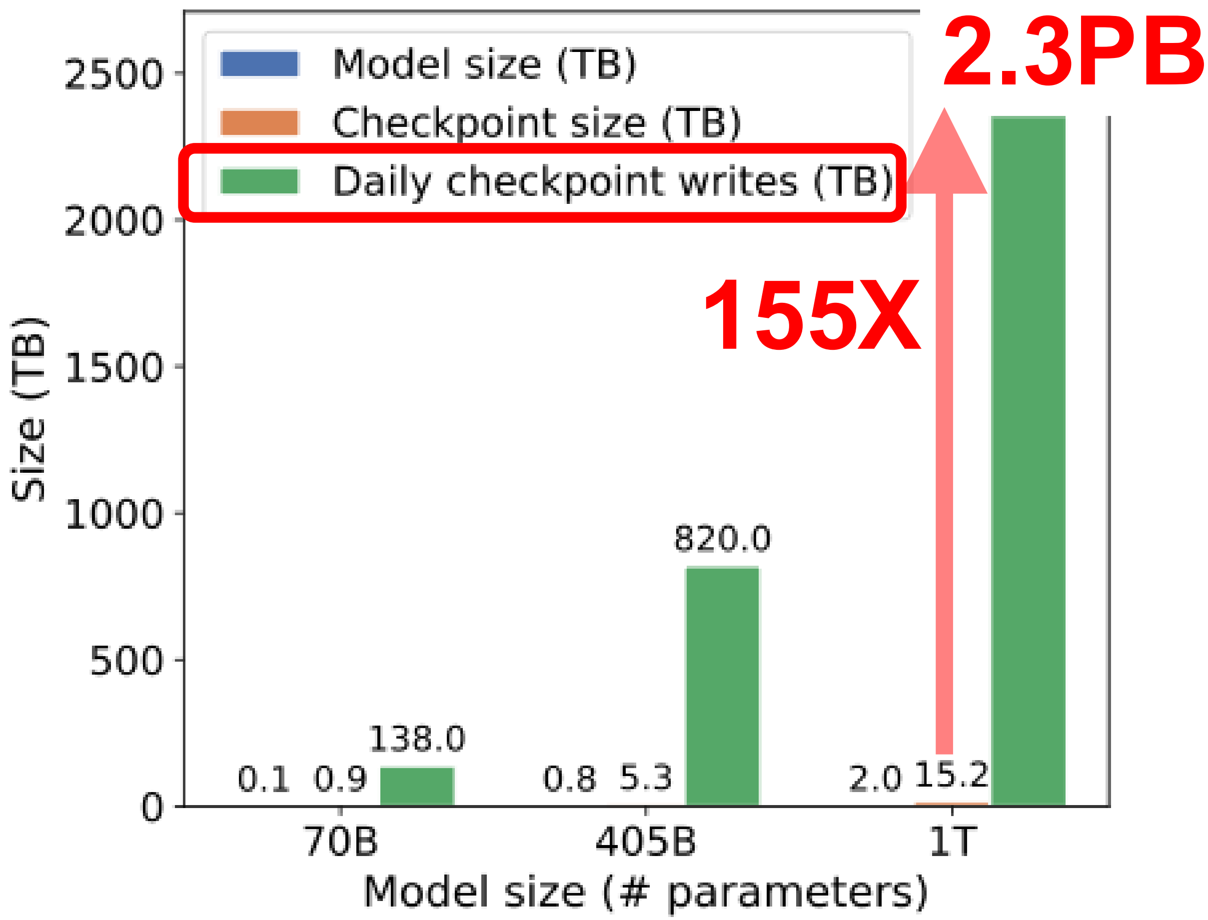


Checkpoint size = model weights + optimizer state

*: https://mlcommons.org/2025/08/storage-2-checkpointing/

LLM training checkpoints are gigantic!

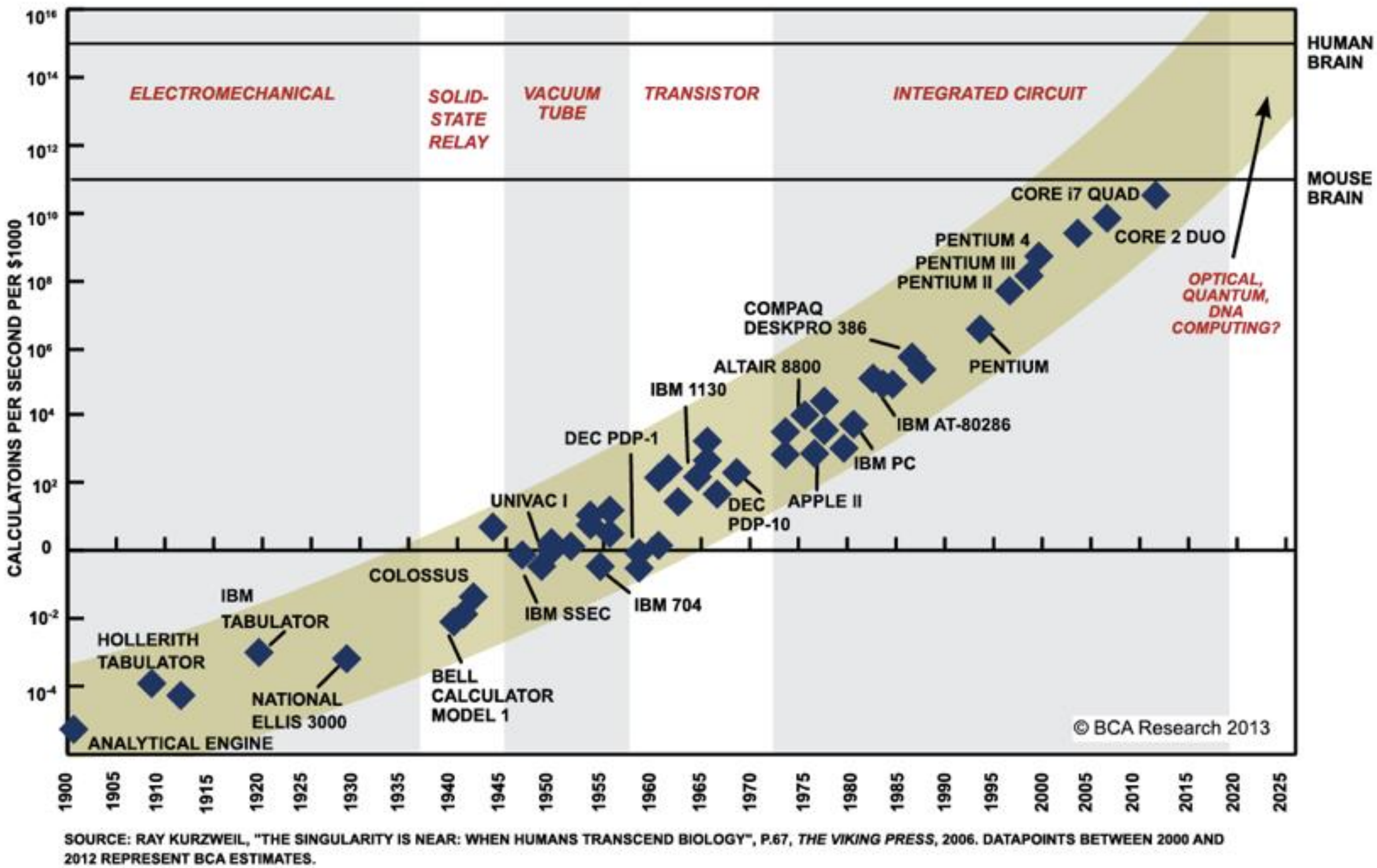
A 16,000-GPU cluster has a MTTF of 186 minutes



A trainer should checkpoint 150+ times a day to mitigate the high cost of frequent failures at 16,000-GPU scale

*: https://mlcommons.org/2025/08/storage-2-checkpointing/

Moore's law is ending

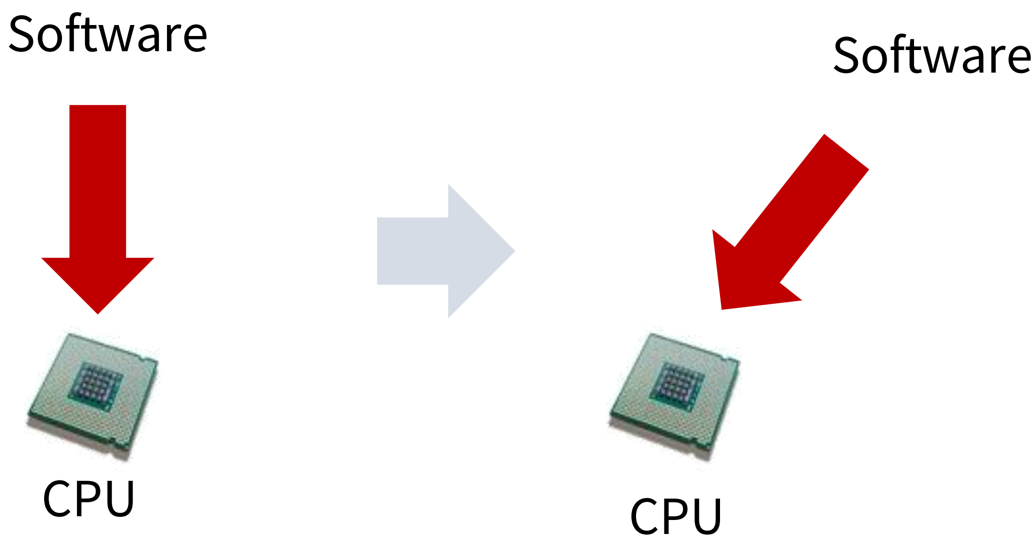


Increased complexity – Computation

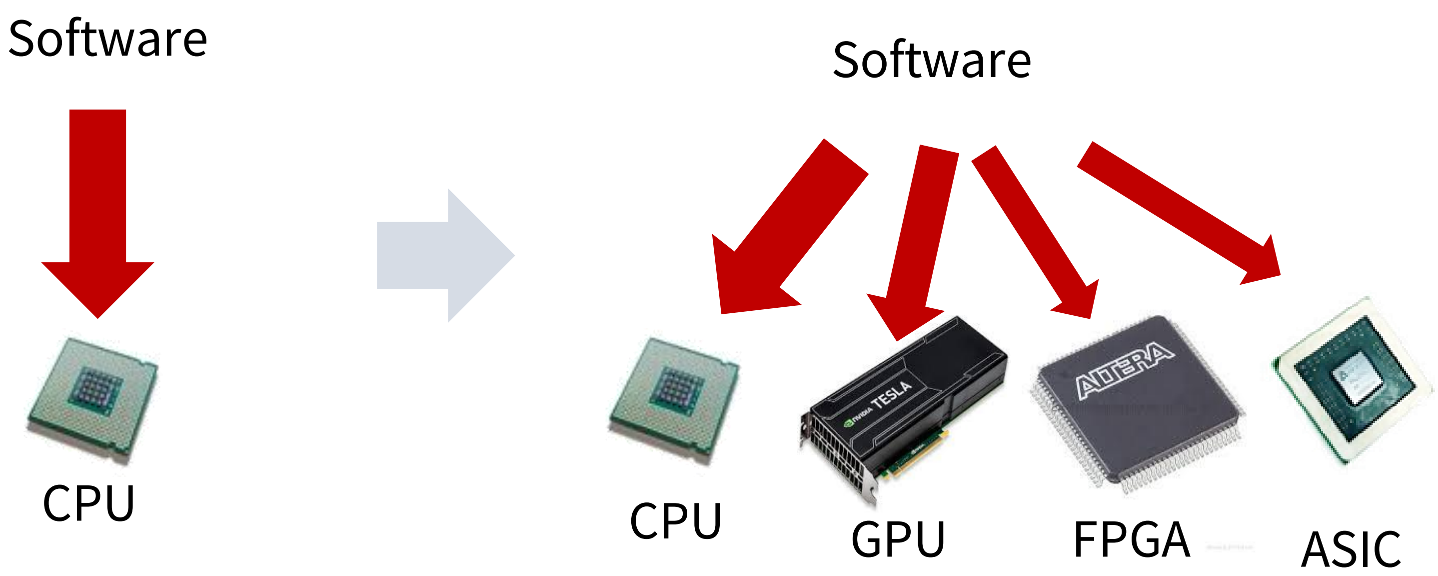
Software

CPU

Increased complexity – Computation



Increased complexity – Computation



Increased complexity – More and more choices in clouds

Basic tier: A0, A1, A2, A3, A4
Optimized Compute : D1, D2, D3, D4, D11, D12, D13
D1v2, D2v2, D3v2, D11v2,…
Latest CPUs: G1, G2, G3, …
Network Optimized: A8, A9
Compute Intensive: A10, A11,…

Microsoft Azure

t2.nano, t2.micro, t2.small
m4.large, m4.xlarge, m4.2xlarge, m4.4xlarge, m3.medium,
c4.large, c4.xlarge, c4.2xlarge,
c3.large, c3.xlarge, c3.4xlarge,
r3.large, r3.xlarge, r3.4xlarge,
i2.2xlarge, i2.4xlarge, d2.xlarge
d2.2xlarge, d2.4xlarge,…

Amazon EC2

n1-standard-1, ns1-standard-2, ns1-standard-4, ns1-standard-8, ns1-standard-16, ns1highmem-2, ns1-highmem-4, ns1-highmem-8, n1-highcpu-2, n1-highcpu-4, n1-highcpu-8, n1-highcpu-16, n1-highcpu-32, f1-micro, g1-small…

Google Cloud

But how do we program this to tackle the challenges of big data in the AI era?

Course syllabus

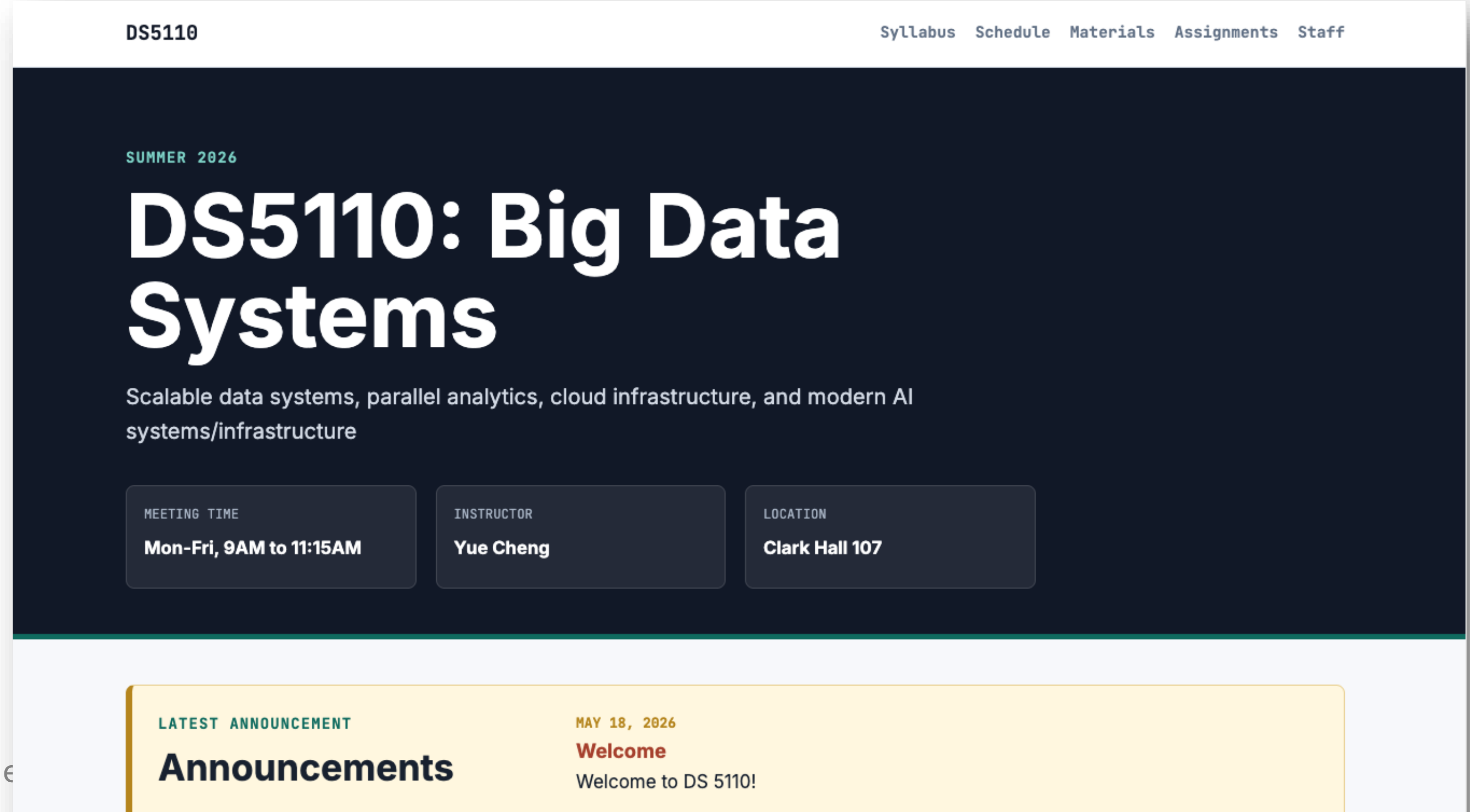
Big picture course goals

- Learn about some of the most influential works in (big) data + AI systems
- Explain the design and architecture
- Read and evaluate some seminal papers
- Develop and deploy applications on open-source data systems and public cloud (AWS, Google Cloud)
- Design and report some data/AI systems ideas

Schedule (tentative)

- Slides, assignments, due dates
- Less concrete further out; don't get too far ahead

https://tddg.github.io/ds5110-summer26/



Lectures (tentative schedule)

- Lecture (+ demos + labs)
    - Slides available on course website (night before)

- Week 1: Basics, MapReduce, Spark + AI tools
- Week 2: Project ideas + cloud data systems
- Week 3: AI systems / infrastructure
- Week 4: Advanced research topics in AI systems + final project presentations

Textbooks?

- Papers, documentations, blog articles (required or optional) serve as reference for many topics that aren't directly covered by a text

- Slides/lecture notes

- Three optional textbooks (first two are free)
  - "**Operating Systems: Three Easy Pieces (OSTEP)**" by Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau
  - "**Distributed Systems (3rd edition)**" by van Steen and Tenenbaum will supply optional alternate explanations
  - "**Designing Data-Intensive Applications (1st edition)**" by Martin Kleppmann (can be accessed via UVA library)

Assignments

- 2.5 hands-on assignments, with **AI coding CLI** (**opencode**), on **AWS**
  - **Assignment 0:** AWS Academy, EC2, Linux shell, and AI coding CLI
  - **Assignment 1:** Analytics with Apache Spark and DuckDB
  - **Assignment 2:** Burst-parallel ML with AWS Lambda

- Short coding assignments
  - Gain hands-on experience with AWS, popular open-source data systems and cloud tools
  - Get exposed to AI coding CLIs

- Assignments and project are all team-based

Grading

- Assignments (50% total)
  - Assignment 0 (10%)
  - Assignment 1 (20%)
  - Assignment 2 (20%)

- Project (30%)

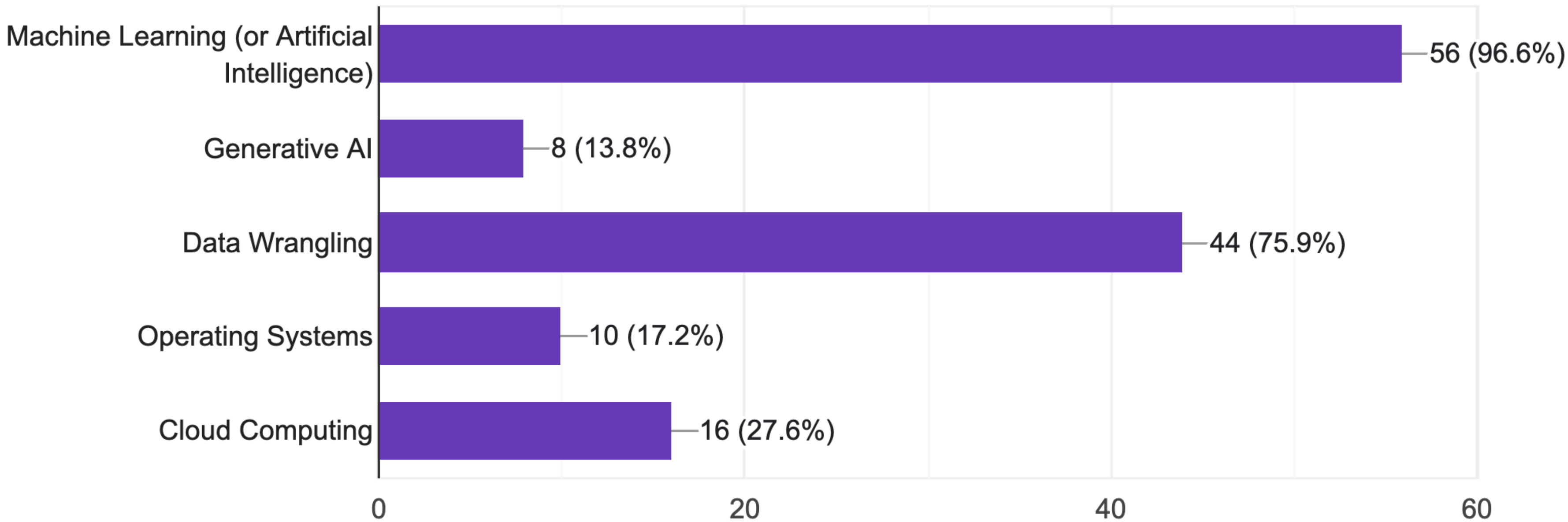
- Final presentation (20%)

Background survey

Background survey

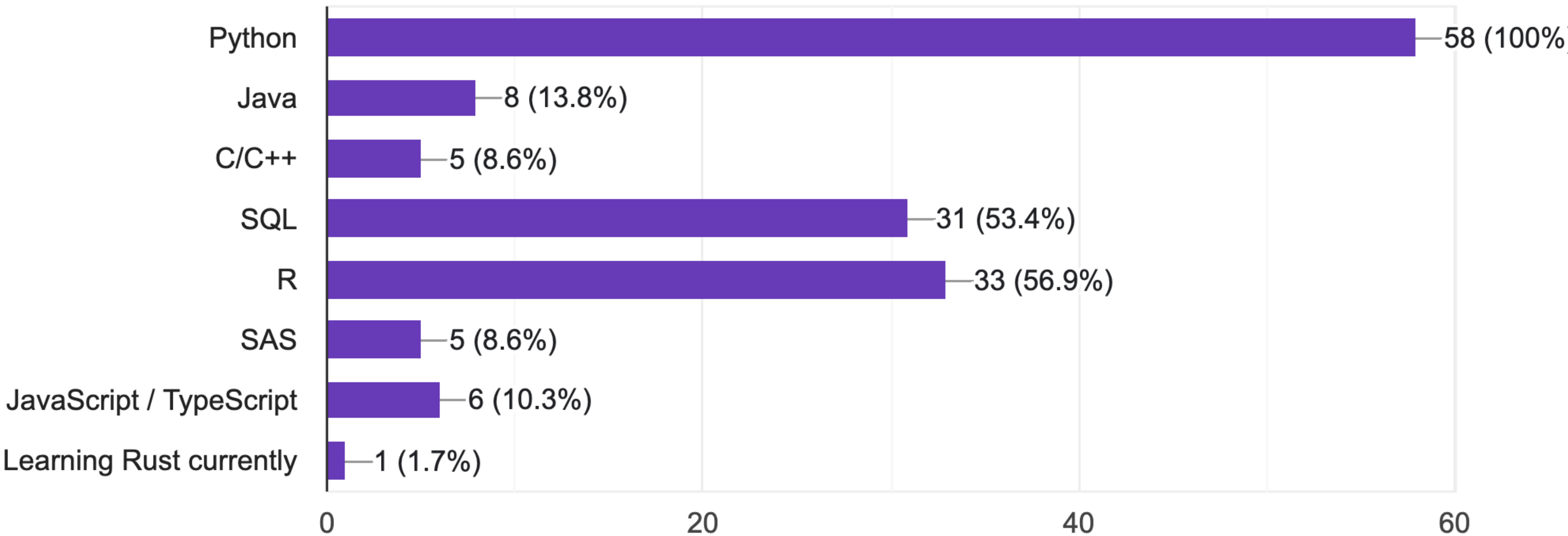
What courses have you taken before

58 responses



Background survey

Which programming languages or query languages are you most comfortable with?
58 responses



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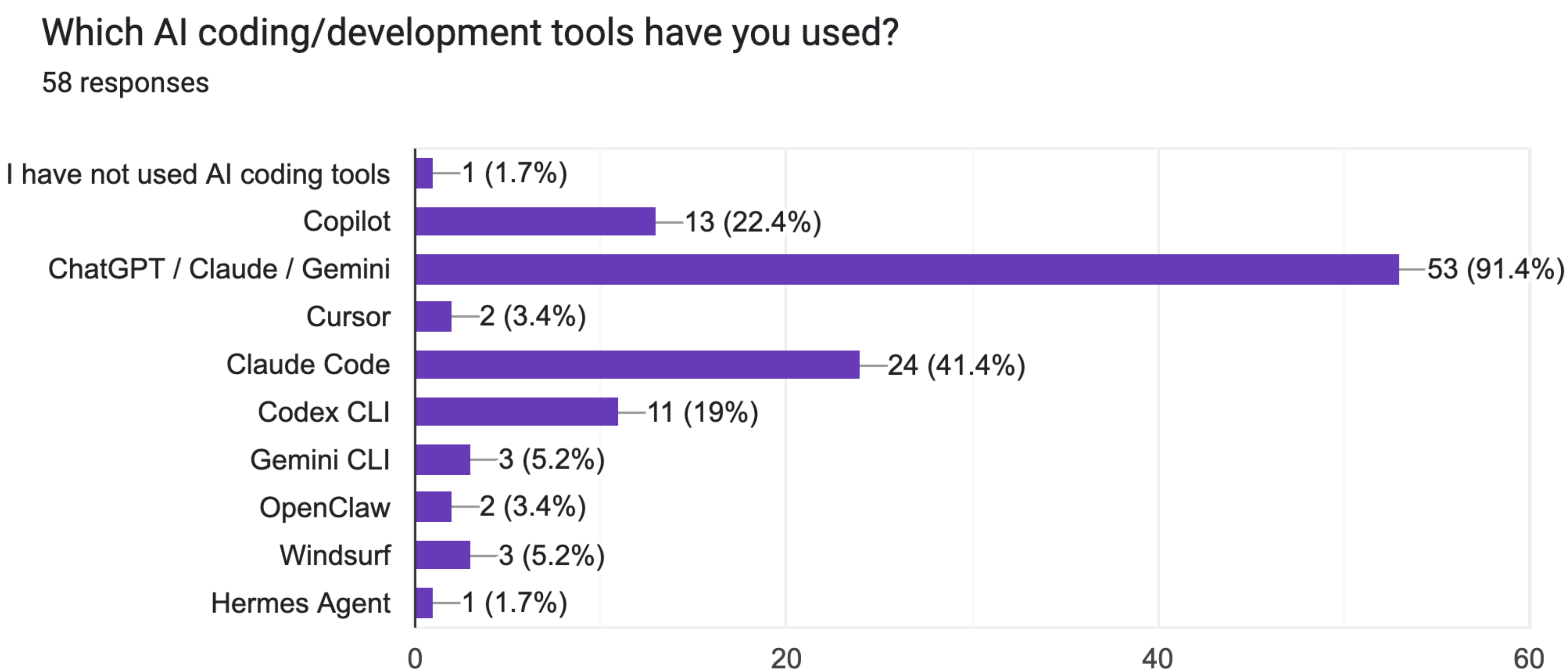
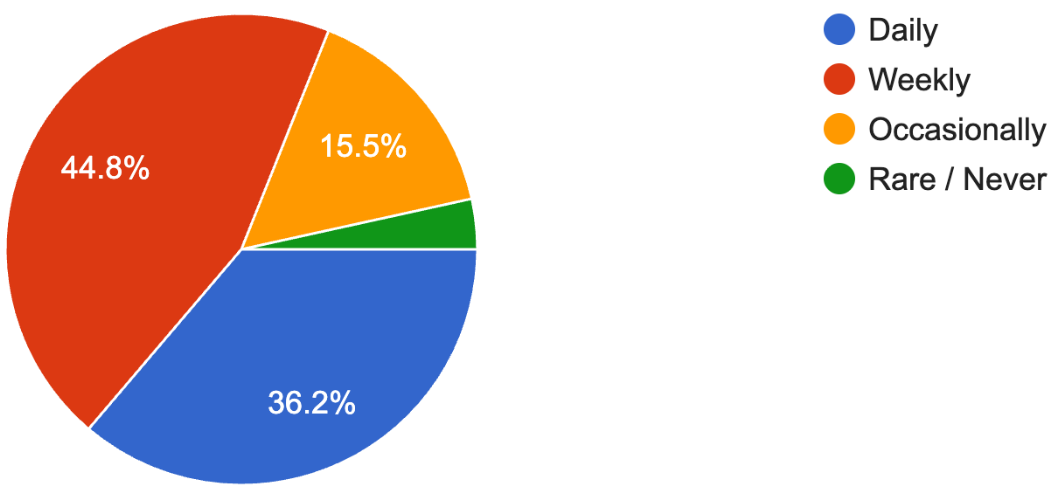


Background survey

How often do you currently use AI coding tools?

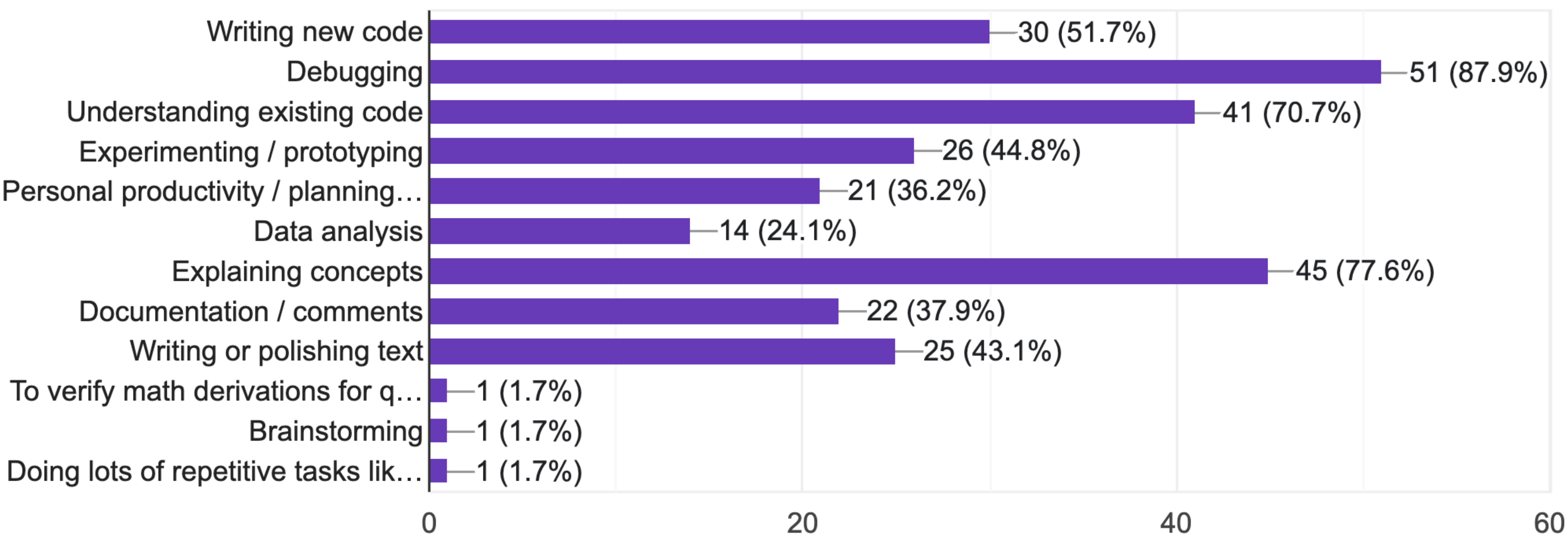
58 responses



Background survey

For which tasks do you primarily use AI tools?

58 responses



Time to introduce yourself